FOR A REGULAR MODULE THE IMAGES OF THE MOMENT MAPS FILL THE VERTEX SPACES EXACTLY WHEN THE MODULE IS ZETA-STABLE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned publisher source sources/algebra-fl01-quiver-varieties-and-branching/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. External references are navigation pointers only; every deliberately kept step of the author's argument stays inside the counted spans. Their printed numbers are the author's own numbering under the author's newtheorem declarations, computed mechanically by the lane's scanner.

Allowed standard prerequisites (editorial).

- The moduli space M_zeta(V,W) of stable representations of the doubled quiver and its regular part, and the moment maps beta_i. Set up in Section 2 of the paper, before the counted statement; the counted statement uses the notation x = [B,a,b] in M_zeta(V,W) from that section. Fixes the

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objects whose images Ima beta_i the statement is about, and the meaning of regularity of the point x.
- The zeta-stability and zeta-semistability of a module (B,a,b) with respect to the stability parameter zeta. Defined in the author's Definition at source lines 746-787 (label def:stable). Gives the property the statement compares with the condition Ima beta_i = V_i for all i in I^0.

Source reference. Original lemma 2.18: sources/algebra-fl01-quiver-varieties-and-branching/pinned-original-source.tex, source lines 1710–1710. Navigation pointer only: the counted proof appeals to this lemma in one line for the forward direction, and the editorially listed standard prerequisites carry the definitions the reader needs; the lemma's own statement and proof are not reproduced here.

Source reference. Original lemma 2.8: sources/algebra-fl01-quiver-varieties-and-branching/pinned-original-source.tex, source lines 999–999. Navigation pointer only: the counted proof uses it once to get zeta-bullet-semistability of the point; the statement itself is not part of the counted material.

Source reference. Original theorem 2.4: sources/algebra-fl01-quiver-varieties-and-branching/pinned-original-source.tex, source lines 853–853. Navigation pointer only: the counted proof takes its Jordan-Holder filtration from this theorem in one sentence; the theorem's statement and proof are not reproduced here.

Lemma 0.1. *Suppose that* $x = [B, a, b] \in \mathfrak{M}_\zeta(V, W)$ *is regular. Then* $\operatorname{Im} \beta_i = V_i$ *for all* $i \in I^0$ *if and only if* (B, a, b) *is* $\zeta^\bullet$*-stable.*

Proof. The 'if' part follows from Lemma 2.18 as the case (a) is excluded as we assumed $W \neq 0$.

The 'only if' part: Since (B, a, b) is $\zeta^\bullet$-semistable by Lemma 2.8 we take its Jordan–Hölder filtration in Theorem 2.4. We must have $k_W = N$, as the submodule V^{kw+1} violates the ζ-stability otherwise. Therefore if $N \neq 0$, $\operatorname{gr}_0(B, a, b)$ has the W-component 0, therefore it is 0-stable module for the subgraph I^0 extended to the whole graph by 0. But the regularity assumption implies that it must be S_i for $i \in I^0$. In particular, β_i is not surjective. □